

ICT2217

Network Security

Lab 3.4 Notes:

AAA Part III – Accounting



Lab Exercise Part 1

1

Configure device access **accounting** using **centralized AAA servers** for storing **logs**.

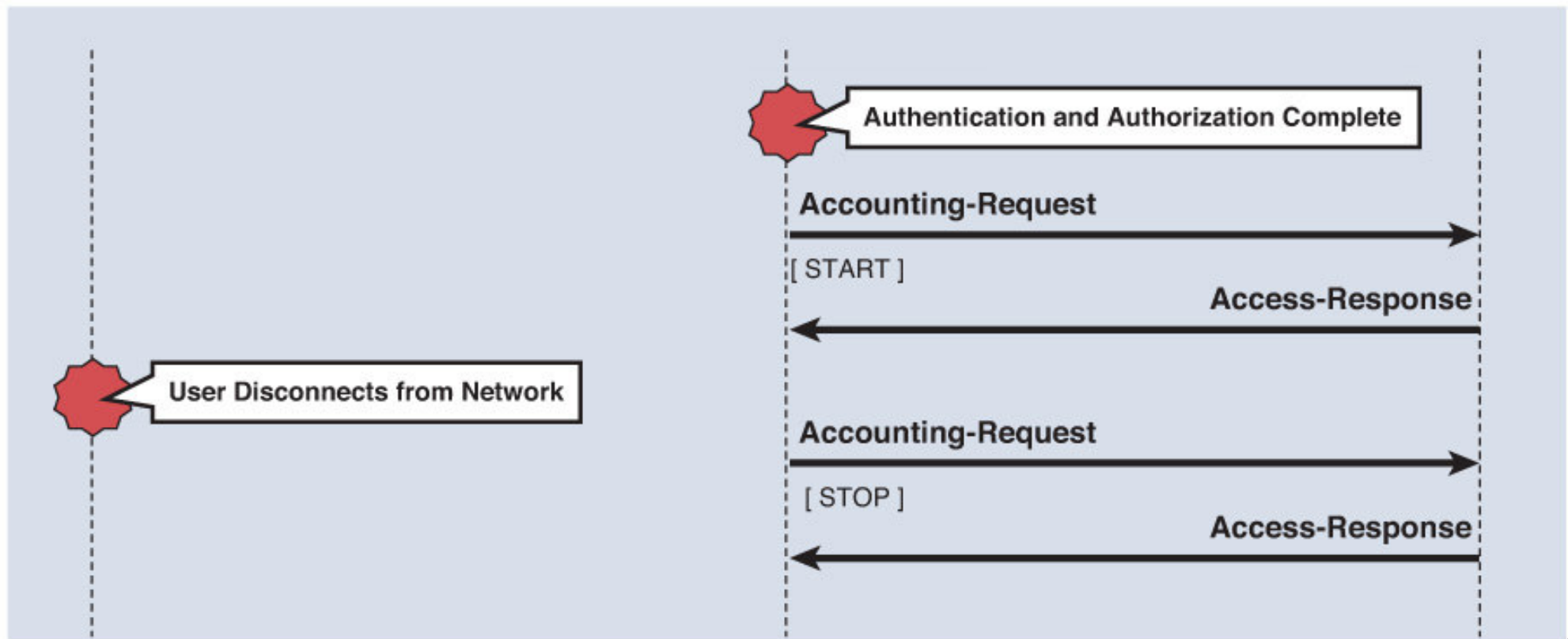
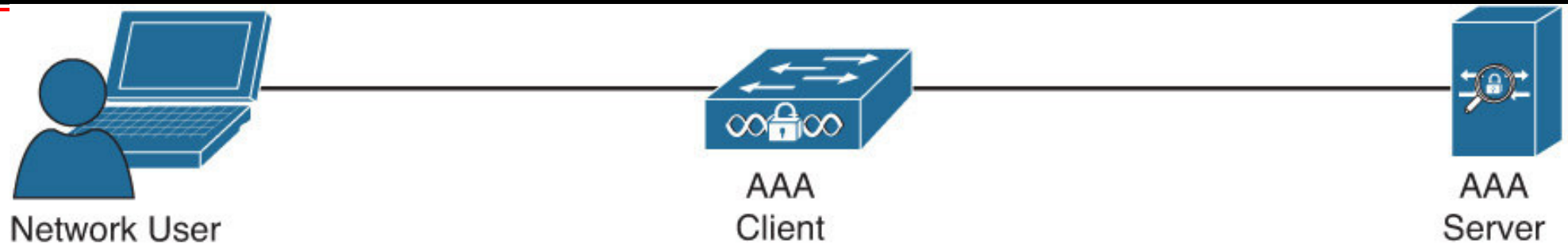
Finally, **AAA accounting** is to have a record of what a user has done which is also crucial to security.

One useful type of accounting is '**exec**' which provides information about user EXEC terminal session after login.

```
R1(config)# aaa new-model
R1(config)# ...
R1(config)# aaa authentication login default group tacacs+
group radius local
R1(config)# aaa authorization exec default group tacacs+
group radius local
R1(config)# aaa accounting exec default start-stop group
tacacs+ group radius
```

- **start-stop**: a **start** message is sent when session begins and a **stop** message is sent when session ends

For **RADIUS**, **accounting** is done separately after authentication and authorization are completed as shown:



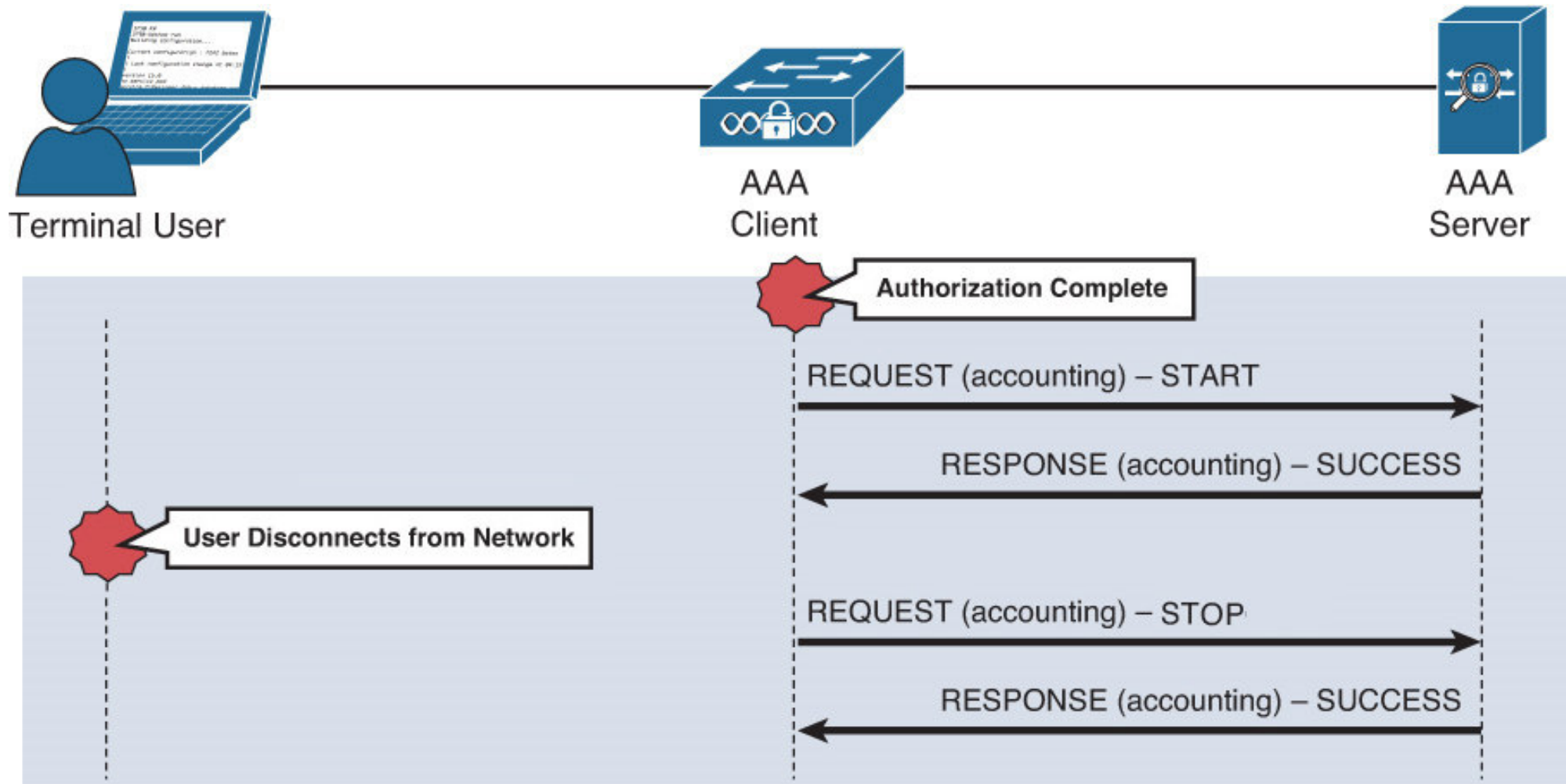
Here's a sample of **exec log** in **freeRADIUS** server which can be found at `/var/log/freeradius/...`

```
Thu Nov  9 13:00:56 2017
Acct-Session-Id = "0000002F"
User-Name = "wendell"
Acct-Authentic = RADIUS
Acct-Status-Type = Start
```

```
NAS-Port = 1
NAS-Port-Id = "tty1"
NAS-Port-Type = Virtual
Service-Type = NAS-Prompt-User
NAS-IP-Address = 192.168.1.200
Acct-Delay-Time = 0
Event-Timestamp = "Nov  9 2017 13:00:56 UTC"
Tmp-String-9 = "ai:"
Acct-Unique-Session-Id = "f1e1339928a412881fcac9c442105e01"
Timestamp = 1510232456
```

```
Thu Nov  9 13:07:28 2017
Acct-Session-Id = "0000002F"
User-Name = "wendell"
Acct-Authentic = RADIUS
Acct-Terminate-Cause = User-Request
Acct-Session-Time = 392
Acct-Status-Type = Stop
NAS-Port = 1
NAS-Port-Id = "tty1"
NAS-Port-Type = Virtual
Service-Type = NAS-Prompt-User
NAS-IP-Address = 192.168.1.200
Acct-Delay-Time = 0
Event-Timestamp = "Nov  9 2017 13:07:28 UTC"
Tmp-String-9 = "ai:"
Acct-Unique-Session-Id = "f1e1339928a412881fcac9c442105e01"
Timestamp = 1510232848
```

Similarly for TACACS+, **accounting** is done separately after authorization as shown:



And here is a sample of **exec log** in **tacacs.net** server which can be found in c:\Users\All Users\TACACS.net\Logs\Accounting_...

A '**start**' message is logged when the exec shell started:

```
<102> 2017-11-09 03:57:41 [192.168.1.200:43972]  
11/09/2017 03:57:41 NAS_IP=192.168.1.200 Port=tty1  
rem_addr=192.168.1.3 User=wendell Flags=Start task_id=25  
timezone=UTC service=shell
```

A '**stop**' message is logged when the exec shell ended:

```
<102> 2017-11-09 03:58:04 [192.168.1.200:32751]  
11/09/2017 03:58:04 NAS_IP=192.168.1.200 Port=tty1  
rem_addr=192.168.1.3 User=wendell Flags=Stop task_id=25  
timezone=UTC service=shell disc-cause=1 disc-cause-ext=9  
pre-session-time=5 elapsed_time=22 stop_time=731022330
```

In addition, **TACACS+** (but not RADIUS) supports another useful type of accounting on **commands** which provide information about specific **privilege level commands** that a user issues.

```
R1(config)# aaa accounting exec default start-stop group  
tacacs+ group radius  
R1(config)# aaa accounting commands 15 default stop-only  
group tacacs+
```

- **stop-only**: a stop message is sent when command is executed

A sample of **commands log** in tacacs.net:

```
<102> 2017-11-09 03:57:47 [192.168.1.200:44251]  
11/09/2017 03:57:47 NAS_IP=192.168.1.200 Port=tty1  
rem_addr=192.168.1.3 User=wendell Flags=Stop task_id=25  
timezone=UTC service=shell priv-lvl=15 cmd=show  
running-config <cr>
```